SPINESURG-CT: A DEEP LEARNING MODEL FOR AUTOMATED SEGMENTATION AND DETECTION OF LUMBOSACRAL TRANSITIONAL VERTEBRAE ON CT IMAGING*

Gregory Schwing, MD, PhD
Department of Surgery (Preliminary)
Detroit Medical Center / Wayne State University
Detroit, MI, USA
gregory.schwing@wayne.edu

Patrick J. Schwing, BS, MS
LSU Health New Orleans School of Medicine
New Orleans, LA 70112, USA

Hazem Eltahawy, MD, PhD
Department of Neurosurgery
Detroit Medical Center
Detroit, MI, USA

ABSTRACT

Purpose: To develop and evaluate SpineSurg-CT, the first CT segmentation model with lumbosacral transitional vertebra (LSTV) awareness built into its architecture — providing voxel-level spatial segmentation of the supernumerary sixth lumbar vertebra (L6) and a simultaneous probabilistic classification of LSTV subtype (Normal / Lumbarization / Sacralization) in a single end-to-end inference pass from raw CT, without requiring upstream vertebra localization or counting.

Materials and Methods: In this retrospective study, 784 CT examinations from the publicly available CTSpine1K COLONOG dataset were used for training (n=552), validation (n=116), and testing (n=116) with stratified sampling ensuring proportional LSTV representation across all partitions. The cohort included 16 lumbarization cases (L6 present), 9 sacralization cases (L5 fused to sacrum), and 759 normal spines. A 3D ResUNet with cross-scale feature fusion modules (CTSpineSegNet architecture) was extended with: (1) a 10-class segmentation output with an explicit L6 class under direct voxel-level supervision; (2) a secondary LSTV classification head at the bottleneck producing calibrated Normal/Lumbarization/Sacralization probabilities; and (3) a multi-strategy class imbalance mitigation stack including novel registration-based sacralization synthesis using ANTs SyN deformation fields computed and applied entirely in preprocessed 256^3 unit coordinate space. Performance was evaluated using Dice similarity coefficient (DSC), 95th-percentile Hausdorff distance (HD95), and mean intersection over union (mIOU), benchmarked against TotalSegmentator.

Results: [SpineSurg-CT achieved mean L6 DSC of X.XX (95% CI: X.XX–X.XX) on lumbarization test cases (n=3), compared with 0.00 for TotalSegmentator. LSTV subtype classification yielded Normal/Lumbarization/Sacralization probabilities with overall accuracy X.XX. Mean L1–L5 DSC was X.XX. All comparisons against TotalSegmentator on LSTV cases were statistically significant (Wilcoxon signed-rank, $p < 0.001$).]

Conclusion: SpineSurg-CT is the first CT segmentation model with LSTV awareness built into its architecture, simultaneously delivering a voxel-level 3D L6 mask and calibrated LSTV subtype probabilities without upstream localization. This represents a clinically meaningful advance for preventing wrong-level spine surgery: existing tools either segment without LSTV awareness (TotalSegmentator), label without segmenting (VERIDAH), or classify without CT spatial output (Kwak et al.).

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***K*eywords** lumbosacral transitional vertebra · spine segmentation · computed tomography · deep learning · wrong-level surgery · L6 · lumbarization · sacralization

Summary Statement: SpineSurg-CT is the first CT segmentation model with lumbosacral transitional vertebra (LSTV) awareness built into its architecture, simultaneously delivering explicit voxel-level L6 segmentation and calibrated Normal/Lumbarization/Sacralization probability output in a single end-to-end pass from raw CT — without requiring upstream vertebra localization or vertebra counting.

Key Results:

1. **[SpineSurg-CT achieved mean L6 DSC of X.XX on lumbarization test cases, compared with 0.00 for TotalSegmentator, which lacks an L6 class by architectural design.]**
2. **[The LSTV classification head produced calibrated Normal/Lumbarization/Sacralization probabilities with F1 of X.XX/X.XX/X.XX — the first probabilistic multi-class LSTV classifier embedded in a CT segmentation architecture, operating without upstream localization or vertebra counting.]**
3. **[Mean L1–L5 DSC was X.XX, comparable to TotalSegmentator (X.XX), with superior lumbosacral junction performance in LSTV cases.]**

Abbreviations: CT, computed tomography; DSC, Dice similarity coefficient; HD95, 95th-percentile Hausdorff distance; LSTV, lumbosacral transitional vertebra; mIOU, mean intersection over union; L6, supernumerary sixth lumbar vertebra; CFFM, cross-scale feature fusion module.

1 Introduction

Wrong-level spine surgery is a sentinel event in neurosurgery, classified as a never-event under the Universal Protocol. Large surgical series consistently report incidence below 1%, but given the volume of lumbar procedures performed annually this represents a substantial preventable burden of harm with severe clinical and medicolegal consequences [1]. Among the anatomical risk factors implicated in vertebral level miscounting, lumbosacral transitional vertebrae (LSTV) are consistently identified as a primary precipitant [1, 2]. In a prospective series of 550 consecutive lumbar surgery patients, Ishida et al. documented LSTV in 12.9% of cases, with vertebral miscounting at initial outpatient evaluation in 54% of LSTV cases and surgical level revision after intraoperative counting in 15% [3].

LSTV encompasses two anatomical subtypes with opposing counting implications. In *lumbarization*, the uppermost sacral segment fails to fuse with the remainder of the sacrum and develops as a supernumerary sixth lumbar vertebra (L6), increasing the mobile segment count by one. In *sacralization*, the fifth lumbar vertebra fuses partially or completely with the sacrum, reducing the mobile segment count by one. LSTV prevalence in the surgical population is 4–18% depending on imaging modality and classification criteria [4, 5].

Computed tomography is the reference standard for LSTV detection and classification. Hanhivaara et al. demonstrated that CT achieves 76% sensitivity and 93% specificity for LSTV, compared with 54% sensitivity for standard lumbar MRI without coronal sequences [6]. Automated CT segmentation could serve as a systematic screening step when CT is available, or as a confirmation tool when LSTV is suspected.

No existing automated CT segmentation system can detect LSTV or label L6. TotalSegmentator — the most widely deployed general-purpose CT segmentation tool, trained on 1,204 CT examinations segmenting 104 anatomical structures — assumes a fixed five-lumbar anatomy and has no L6 class by design [7]. VERIDAH [8] is a sophisticated vertebra labeling algorithm achieving 97.22% lumbar enumeration anomaly recall on CT. It uses a 3D DenseNet169 with four learned classification heads (vertebra label, anatomical region, transition type, visibility) and a constrained sequence optimizer — it does learn morphological features from CT. However, L6 is represented *implicitly*: during training, L6 labels are replaced with two consecutive L5 labels, and the sequence optimizer detects lumbarization by finding that two consecutive L5-class predictions minimise the global path cost. There is no explicit L6 class, no direct voxel-level L6 supervision, and no 3D spatial mask. VERIDAH outputs vertebra centroid labels and requires pre-computed vertebra localizations from an upstream segmentation model (SPINEPS), precluding end-to-end inference from raw CT

Critically, Schinz et al. recently demonstrated at the *thoracolumbar* junction that morphometric vertebral shape-based classification outperforms counting-based approaches on all metrics — achieving 100% agreement with nerve morphology (the functional ground truth) versus 92.6% for the counting-based standard — and is robust to incomplete spine imaging where counting fails entirely [9]. This validates the principle that morphological feature learning is superior to counting for vertebral transition detection. However, Schinz et al. provide a manual classification scheme

applied by expert radiologists; no automated neural network model for this task exists, and the lumbosacral junction — clinically more consequential for wrong-level surgery — was explicitly not assessed.

A ResNet-50 classifier operating on plain radiographs achieves 76.4% accuracy for binary LSTV detection but provides no segmentation, no subtype distinction, and requires a separate imaging modality [10].

We therefore developed SpineSurg-CT, extending CTSpineSegNet [11] with three novel components for LSTV detection: (1) a 10-class segmentation output including an explicit L6 class, (2) a secondary multi-task LSTV classification head predicting Normal/Lumbarization/Sacralization from learned morphological features rather than counting, and (3) a multi-strategy class imbalance mitigation stack including registration-based sacralization synthesis. To our knowledge, SpineSurg-CT is the first CT segmentation model with an explicit L6 class under direct voxel-level supervision, producing a 3D spatial segmentation mask without upstream localization steps. Whereas VERIDAH detects lumbarization through implicit sequence optimization over relabelled L5 centroids, SpineSurg-CT learns L6 morphology directly from annotated voxels, extending the morphometric classification principle validated by Schinz et al. at the thoracolumbar junction [9] to a fully automated spatial segmentation framework at the lumbosacral junction.

2 Materials and Methods

2.1 Study Design

This retrospective study used publicly available datasets. Institutional review board approval and informed consent were not required. The study was conducted and reported in accordance with the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) 2024 guidelines [12].

2.2 Datasets

2.2.1 CTSpine1K — Primary Training Dataset

CTSpine1K [13] is a publicly available vertebral segmentation benchmark comprising 1,005 CT examinations across four clinical subsets. The COLONOG subset (784 cases, colonoscopy screening CTs) was selected as the appropriate training cohort for lumbosacral segmentation: all cases include complete L1–L5 lumbar coverage in the supine position. The COVID-19 and HNSCC-3DCT subsets were excluded due to truncated field of view, confirmed by absence of complete lumbar labels in a pre-training audit.

Within the 784 COLONOG cases, LSTV prevalence was determined by systematic label scanning: 16 cases contained raw label 25 (L6, lumbarization) and 9 cases had raw label 24 (L5) absent with remaining lumbar labels present (sacralization), for 25 total LSTV cases (3.2% prevalence). A unified 10-class label mapping was applied: background (0), L1–L5 (1–5), L6 (6), sacrum (7), left hip (8), right hip (9).

2.2.2 CTPelvic1K — Supplementary Pelvic Supervision

CTPelvic1K [14] Dataset 1 contains pelvic masks for 35 COLONOG cases. Datasets 2–7 originate from independent clinical studies with no patient overlap with CTSpine1K COLONOG, confirmed by DICOM UID analysis. Patient-number-based matching of Dataset 1 to CTSpine1K identified 33/35 matched pairs (94% match rate). No LSTV cases overlap with the 33 matched pelvic cases (zero intersection confirmed empirically). The 33 matched cases provide sacrum (class 7), left hip (class 8), and right hip (class 9) ground truth via online nearest-neighbour resampling into spine CT space during training. Critically, these 33 cases are the only samples in the dataset with complete lumbosacral label coverage including the pelvic girdle; the remaining 751 cases carry lumbar vertebra labels only.

2.3 Dataset Split

Dataset partitioning used stratified sampling across three priority-ordered strata: (1) LSTV cases (n=25); (2) normal spine cases with matched pelvic mask (n=33); (3) normal spine-only cases (n=726). Each stratum was independently shuffled (seed = 42) and split 70/15/15, yielding: train n=552, validation n=116, test n=116. This guarantees a minimum of 1 LSTV case in both validation and test sets, and ensures that pelvic mask cases appear in all three partitions. Table 1 summarises dataset characteristics.

2.4 Image Preprocessing

All images were reoriented to PIR convention (posterior-inferior-right axial orientation, per **(author?)** 11), resampled to 1 mm isotropic spacing (bilinear for images, nearest-neighbour for labels), HU-clipped to $[-1000, 1000]$ and linearly

Table 1: Dataset Characteristics by Stratum and Partition

Stratum	Total	Train	Val	Test	LSTV
LSTV cases	25	17	4	4	25 (100%)
Lumbarization (L6 present)	16	11	2	3	—
Sacralization (L5 fused)	9	7	1	1	—
Spine + pelvic mask	33	23	5	5	0
Spine only	726	512	107	107	0
Total	**784**	**552**	**116**	**116**	25 (3.2%)

LSTV, lumbosacral transitional vertebra. Spine + pelvic mask cases have confirmed zero overlap with LSTV cases. Stratified 70/15/15 split applied independently within each stratum.

normalised to $[0, 1]$, foreground-cropped based on image intensity content (20-voxel margin), zero-padded if necessary, and centre-cropped to 256^3 voxels.

Image-intensity-based foreground cropping — rather than label-based cropping — was used deliberately to preserve the full lumbosacral and pelvic anatomy within the 256^3 field of view. Label-based cropping computes the bounding box from the spine labels only, which terminate at the inferior sacrum; the hip bones (classes 8–9) extend 40–60 mm below this boundary and would be systematically excluded before pelvic mask fusion, rendering pelvic ground truth absent from training despite being available. Image-based cropping captures the full bony extent of the body in the field of view, ensuring sacrum and hip labels are physically present in the training volume after `FuseSpinePelvicLabels`. For cases without a matched pelvic mask, the additional inferior region contains only background-class air voxels (intensity ≈ 0 after normalisation), which are indistinguishable from zero-padding and impose no learning cost. This single preprocessing function is shared identically between training, registration-cache computation, and 3D augmentation visualisation, guaranteeing that what is visualised in the quality-control output is exactly what the network receives during training. The resulting volumes occupy a shared unit coordinate space (origin at zero, 1 mm isotropic spacing, identity direction cosines) exploited by the registration-based augmentation pipeline described in Section 2.6.

2.5 Model Architecture

SpineSurg-CT extends CTSpineSegNet [11], a 3D residual U-Net with cross-scale feature fusion modules (CFFM) at encoder skip connections between layers $1\to2$ and $2\to3$, and a linear self-attention bottleneck reducing complexity from $O(n^2)$ to $O(kn)$ with $k{=}64$. Three extensions were added:

1. **10-class segmentation output:** L6 (class 6) added with a $20\times$ loss weight in the combined Dice-CE objective to compensate for extreme rarity (16/784 cases, 2.0%).
2. **LSTV classification head:** Secondary branch at the bottleneck: global average pooling $\to$ FC$(512\to128\to3)$ with ReLU and 0.3 dropout, producing Normal/Lumbarization/Sacralization probabilities (loss weight $\alpha{=}0.5$).
3. **Data augmentation:** Random flip ($p{=}0.5$, axis 0), rotation $\pm30^\circ$ ($p{=}0.5$, all axes), zoom 0.7–$1.4\times$ ($p{=}0.5$), Gaussian noise $\sigma{\leq}0.1$ ($p{=}0.3$), contrast jitter $\gamma{=}0.7$–1.5 ($p{=}0.3$), intensity scale/shift $\pm10\%$ ($p{=}0.3$ each).

Total trainable parameters: 15,877,165.

2.6 Class Imbalance Mitigation

Four strategies addressed LSTV class imbalance (3.2% prevalence):

1. WeightedRandomSampler. LSTV cases oversampled at $10\times$, ensuring approximately one LSTV case per batch at batch size 2.

2. Class-weighted loss. Per-class loss weights: $20\times$ (L6), $2\times$ (sacrum, hips), $1\times$ (L1–L5), $0.1\times$ (background). Sacralization samples receive an additional $2\times$ sacrum weight boost applied per-sample in the cross-entropy term, reflecting the anatomically enlarged sacrum produced by L5 incorporation.

3. CopyPaste lumbarization synthesis with targeted elastic deformation. The 11 lumbarization training cases were preprocessed offline into an in-memory pool. During training, 30% of normal spine samples received synthetic lumbarization as follows. A random pool case was selected and its L6 label bounding box extracted with a 4-voxel margin. In lumbarization, L6 is the supernumerary segment that interposes between L5 and the sacrum — it is *inferior*

to L5, occupying the L5–S1 disc space. Accordingly, the L6 patch was placed at the *inferior* boundary of the target's L5 label along the inferior-to-superior axis (axis 1 in PIR space): the patch was positioned such that its superior face aligns with the inferior edge of target L5, reproducing the anatomical relationship. Along the posterior-to-anterior axis (axis 0) and right-to-left axis (axis 2), the patch was centred on the target spine column centroid. Image intensities and label values were updated simultaneously within the L6 mask region. Elastic pre-deformation of the source patch was deliberately omitted: deforming the source before bounding box extraction scatters label voxels outside the pre-computed box, producing floating label fragments disconnected from the spine column.

Following the paste, targeted elastic deformation was applied exclusively to the L5–L6–sacrum subvolume (bounding box of classes 5, 6, 7 with 24-voxel context margin), leaving L1–L4 unperturbed. This is anatomically appropriate: in lumbarization, the upper lumbar spine is completely normal and elastic deformation of L1–L4 would introduce spurious morphological variation into the supervision signal for these classes. The 24-voxel margin provides sufficient spatial context to prevent displacement field discontinuities at the L4/L5 boundary without deforming L4 itself. The identical displacement field was applied to both image and label simultaneously using MONAI `Rand3DElasticD`: bilinear interpolation for image intensities and nearest-neighbour interpolation for the discrete label volume. The label requires no boundary blending — discrete segmentation masks have no intensity seam artefact — while image intensities at the subvolume boundary were smoothly composited with the original using a Gaussian-feathered blend mask (linear fade over 24 voxels per face) to eliminate visible intensity discontinuities at the elastic subvolume boundary.

4. Registration-based sacralization synthesis. This is the methodologically novel contribution of the present work. The core challenge in sacralization augmentation is that simple label modification (zeroing the L5 annotation) leaves CT image intensities unchanged, producing supervision signal without a corresponding image-domain deformation. A realistic synthetic sacralization case must simultaneously exhibit: collapsed L5–S1 disc space, enlarged transverse processes, and superior elongation of the sacrum.

Phase 1 — offline registration (one-time cost, ≈45 min). For each of the 7 sacralization cases in the training split, the CT volume was preprocessed to 256^3 1 mm isotropic and registered to a reference normal spine using ANTs SyNRA deformable registration [15]. Critically, both registration and subsequent field application were performed entirely in the shared preprocessed unit coordinate space (PIR, 256^3, 1 mm isotropic, unit origin and direction), not in the original NIfTI physical space. This eliminates the coordinate frame mismatch that arises when fields computed in heterogeneous native NIfTI spaces are applied to standardised preprocessed tensors. Reference normals were drawn exclusively from the 23 pelvic-masked training cases, ensuring that the deformation target has complete sacrum and hip labels (classes 7–9). The forward deformation field — encoding the geometric transformation from normal to sacralization morphology — was cached to disk alongside preprocessed volume arrays in NumPy format for instant retrieval on subsequent runs.

Phase 2 — online synthesis (per-batch, ≈2 s). Unlike the CopyPaste augmentation, which applies targeted local elastic deformation restricted to the L5–L6–sacrum subvolume, sacralization synthesis applies the ANTs deformation field to the entire 256^3 volume. This is appropriate: the registration field was computed from a globally coherent morphological difference between a real sacralization case and a normal reference, encoding simultaneous changes across the entire lumbosacral anatomy — caudal migration of L5 toward the sacrum, collapse of the L5–S1 disc space, enlarging transverse processes, and superior elongation of the sacrum. Truncating this field to a subvolume would arbitrarily corrupt a globally consistent deformation and introduce artificial discontinuities absent in the original anatomy. During training, a matched normal case (one with a confirmed pelvic mask, n=23 in training) was selected as the augmentation target. The Boolean flag `has_pelvic_mask` was propagated through the preprocessing pipeline to guarantee that field application is restricted to cases with complete label coverage, regardless of MONAI tensor metadata state. A randomly selected cached field was applied to both image and label tensors simultaneously using ANTs linear interpolation (image) and nearest-neighbour (label). Post-warp L5 voxels geometrically adjacent to the sacrum — identified by binary dilation of the sacrum mask with 8-voxel structuring element (8 mm at 1 mm isotropic) — were relabelled as sacrum (class 7), modelling L5 incorporation. If more than 50% of L5 was found adjacent to the sacrum, all remaining L5 voxels were absorbed (complete sacralization). The synthetic label thus contains a fused L5+sacrum structure with intact hip labels, reflecting realistic sacralization anatomy. Because all 23 matched training cases are eligible and augmentation probability is set to 1.0, approximately 23 anatomically complete synthetic sacralization samples are generated per training epoch — the hard ceiling imposed by the number of pelvic-masked training cases in this dataset. The ANTs deformation field was computed from spine anatomy only; its magnitude decays smoothly to near-zero in the inferior pelvis region where no corresponding spine structure drives the registration, so hip label integrity in the synthetic output is preserved without explicit masking or protection.

2.7 Training Configuration

Training used AdamW (lr=10^{-3}, weight decay 10^{-5}), linear warmup (5 epochs) then cosine annealing to $\eta_{\min}=10^{-6}$, maximum 300 epochs with early stopping (patience 30, monitor: `val/DSC`, $\Delta_{\min}=10^{-4}$), batch size 2, BF16 mixed precision, gradient clipping 1.0. Hardware: NVIDIA H200 NVL (141 GB VRAM, Hopper architecture). `torch.set_float32_matmul_precision('high')` was set to exploit native BF16 tensor cores. Framework: PyTorch 2.4.1 (CUDA 12.4), PyTorch Lightning 2.4.0, MONAI 1.4.0, antspyx (ANTs Python bindings).

2.8 Evaluation

Primary evaluation compared SpineSurg-CT against TotalSegmentator (v2.4.0, CT model) on the 116-case held-out test set. Metrics: DSC, HD95 (mm), and mIOU per class. LSTV classification: sensitivity, specificity, F1, and AUROC per class. Statistical significance: Wilcoxon signed-rank test, $p<0.05$. Confidence intervals: bootstrap resampling ($n=1000$). The pre-specified **primary outcome** is L6 DSC on lumbarization test cases; TotalSegmentator L6 DSC is 0.00 by design.

3 Results

3.1 Segmentation Performance

[Table 2 reports per-class DSC, HD95, and mIOU for SpineSurg-CT and TotalSegmentator on the 116-case test set. SpineSurg-CT achieved mean L1–L4 DSC of X.XX ± X.XX, comparable to TotalSegmentator (X.XX ± X.XX, p=X.XX).]

Table 2: Segmentation Performance on Held-Out Test Set (n=116)

	DSC		HD95 (mm)		mIOU	
Class	SpineSurg	TotalSeg	SpineSurg	TotalSeg	SpineSurg	TotalSeg
L1	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L2	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L3	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L4	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L5	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
L6†	[X.XX]	0.00	[X.XX]	N/A	[X.XX]	N/A
Sacrum	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Left hip	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Right hip	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]	[X.XX]

†L6 restricted to lumbarization test cases (n=3). TotalSegmentator L6 DSC = 0.00 by architectural design (no L6 class). N/A, not applicable.

3.2 Lumbarization Detection — Primary Outcome

[On the 3 lumbarization test cases, SpineSurg-CT achieved mean L6 DSC of X.XX (95% CI: X.XX–X.XX), HD95 of X.XX mm. TotalSegmentator achieved L6 DSC = 0.00 on all 3 cases ($p<0.001$). Figure 5 shows a representative segmentation.]

3.3 LSTV Classification

[Table 3 summarises LSTV classification performance. Overall accuracy was X.XX.]

4 Discussion

SpineSurg-CT is the first CT segmentation model with explicit L6 supervision and multi-class LSTV detection. The central finding is that SpineSurg-CT achieves positive L6 DSC on lumbarization cases where TotalSegmentator scores 0.00 by architectural necessity. This is a capability gap, not a performance margin.

Table 3: LSTV Classification Performance

System	Input	Sensitivity	Specificity	F1	Subtype output
SpineSurg-CT (this work)	CT	**[X.XX]**	**[X.XX]**	**[X.XX]**	Lumb / Sacral / Normal
Kwak et al. [10]	Radiograph	0.851	0.619	N/A	Binary

Kwak et al. comparison is binary (LSTV/Normal) on plain radiograph; SpineSurg-CT provides three-class CT output. Direct comparison should be interpreted cautiously given different modality and granularity.

The clinical motivation is direct. LSTV affects 4–18% of the general population [4, 5] and causes systematic vertebral miscounting, the proximate cause of wrong-level spine surgery. Ishida et al. documented surgical level revision in 15% of LSTV surgery cases [3]; Albano et al. identified LSTV as the primary anatomical risk factor in a systematic review of wrong-level surgery [1]. SpineSurg-CT removes the barrier of missing L6 representation in all existing automated CT tools.

The comparison with VERIDAH [8] merits precise framing. VERIDAH uses a 3D DenseNet with four learned classification heads and a constrained sequence optimizer — it does learn morphological features from CT and is substantially more than a simple counting system. The key architectural distinction from SpineSurg-CT is that VERIDAH represents L6 *implicitly*: during training, L6 labels are replaced with two consecutive L5 labels, and the sequence optimizer detects lumbarization by finding that two consecutive L5 predictions minimise the global path cost. Consequently, there is no explicit L6 class, no direct voxel-level L6 supervision, and no 3D spatial output — VERIDAH produces vertebra centroid labels and requires pre-computed localizations from an upstream segmentation model.

SpineSurg-CT is complementary: L6 is an explicit segmentation class learned directly from annotated voxels, producing a 3D mask of the supernumerary segment usable for surgical planning in a single end-to-end inference pass from raw CT. Schinz et al. demonstrated at the thoracolumbar junction that explicit shape-based classification achieves perfect agreement with functional anatomy (nerve morphology, 100% vs. 92.6% for counting-based methods), while counting-based classification produced discrepant labels in 13.3% of cases [9]. Although Schinz et al. addressed a different junction, the principle that explicit morphological feature supervision produces more reliable transitional vertebra detection applies to the lumbosacral problem. The practical difference between VERIDAH and SpineSurg-CT is therefore implicit sequence-level L6 detection with centroid output versus explicit voxel-supervised L6 segmentation with a 3D spatial mask and probabilistic LSTV classification.

The registration-based sacralization synthesis pipeline represents a novel methodological contribution. By performing ANTs SyN deformable registration and field application entirely within the shared preprocessed 256^3 unit coordinate space, we eliminate the coordinate frame mismatch that arises when fields computed in heterogeneous native NIfTI physical spaces are applied to standardised training tensors. The resulting synthetic cases exhibit simultaneous image and label deformation: the collapsed L5–S1 disc space, enlarged transverse processes, and elongated sacrum morphology that characterise radiological sacralization are present in the CT intensities, not merely in the annotation. This is a qualitative advance over label-deletion approaches, which remove the L5 annotation while leaving the underlying CT anatomy unchanged. Restricting field application to the 23 training cases with complete pelvic ground truth (sacrum and hip labels) ensures that every synthetic sacralization sample contributes anatomically complete lumbosacral supervision signal, including a fused L5+sacrum structure with intact hip labels.

4.1 Limitations

Dataset size and LSTV case count. The training cohort of 784 cases with 25 LSTV cases is small relative to general-purpose segmentation benchmarks. L6 DSC estimation on 3 lumbarization test cases and LSTV classification on 4 test cases has high variance; confidence intervals should be interpreted accordingly and results treated as proof-of-concept pending external validation.

Synthetic sacralization ceiling. The matched-only restriction on sacralization synthesis — necessary to produce anatomically complete sacrum and hip labels — limits augmentation to the 23 pelvic-masked training cases, yielding approximately 23 synthetic sacralization samples per epoch. This represents the hard ceiling imposed by the intersection of the CTSpine1K COLONOG and CTPelvic1K Dataset 1 cohorts. The model learns sacralized sacrum morphology primarily from these 23 cases plus the 7 real sacralization training cases, with limited diversity relative to the lumbarization augmentation pipeline which can operate on all 510 normal training cases.

Deformed-normal-sacrum approximation. Synthetic sacralization is generated by warping a morphologically normal matched reference toward the sacralization appearance template. The resulting sacrum label covers the fused L5+sacrum territory but reflects the geometry of a deformed normal sacrum rather than the true bony morphology of sacralization,

in which the L5 body is incorporated into the superior sacral ala with altered trabecular architecture, cortical thickening, and transverse process hypertrophy. The model may therefore underestimate sacrum boundary precision at the L5 fusion zone. Sacralization sacrum DSC on test cases should be interpreted with this approximation in mind.

No pelvic ground truth for real sacralization training cases. The 7 sacralization training cases have no matched CTPelvic1K pelvic mask, as the CTSpine1K LSTV cases do not overlap with the CTPelvic1K COLONOG subset. Consequently, sacrum and hip supervision for these cases derives entirely from synthetic augmentation of pelvic-masked normals.

Deformation field applied to sacrum label in synthetic sacralization. A more principled sacralization synthesis would warp only the lumbar vertebra labels (L1–L5) of the normal reference toward the sacralization template while holding the sacrum label fixed as the stationary fusion target. This would ensure that the sacrum label remains correctly anchored to the underlying CT anatomy and that the dilation-based L5 fusion step always operates against a geometrically consistent sacrum boundary. In the current implementation, `ants.apply_transforms` is applied to the full label volume including the sacrum class, which introduces a small but non-zero distortion of the sacrum label relative to its true position. Although the 8-mm binary dilation used in the fusion step provides some tolerance for this drift, the distortion may reduce sacrum boundary precision at the L5 fusion zone.

Implementing the asymmetric approach requires knowledge of which voxels constitute the sacrum in both the sacralization template images and the normal reference volumes. For normal reference volumes this information is available via the CTPelvic1K matched masks — which is why synthesis is restricted to the 23 pelvic-masked training cases. For the 7 real sacralization training cases, however, CTSpine1K provides neither an L5 label (anatomically fused and therefore unlabelled) nor a sacrum mask (no matched pelvic case exists). This creates a fundamental labelling gap: it is not possible to separate the sacralized L5 from the native sacrum in these cases without either (a) manual expert annotation of the L5–S1 fusion boundary, which is radiologically challenging and subjective even for experienced spine radiologists, or (b) pseudo-labelling using TotalSegmentator [7], which segments sacrum and hips by morphological priors independently of vertebral counting and would correctly identify the pelvic structures on all 784 cases. Pseudo-label quality on the sacralization cases should be verified against the 33 manually annotated matched cases before adoption. Either approach would enable the sacrum-fixed deformation strategy described above and represents a concrete improvement target for a subsequent study (see Section 4.2).

No Castellvi subtype classification. SpineSurg-CT detects LSTV type but does not provide Castellvi subtype (Types I–IV). This requires morphometric analysis of transverse process height and sacral contact distance, planned as a second-stage model.

No explicit fused L5–S1 segmentation. In sacralization, the fused L5 is incorporated into the sacrum label following CTSpine1K annotation convention. The model does not distinguish the sacralized L5 from the native sacrum within class 7.

CT only; single-centre training data. All training data originates from COLONOG colonoscopy screening CTs acquired in a single imaging protocol. External validation on diverse clinical CT acquisitions — including different scanners, reconstruction kernels, and patient populations — is required before deployment. MRI LSTV detection remains a separate problem addressed for the labelling task by VERIDAH [8].

4.2 Future Directions

CycleGAN domain adaptation for sacralization synthesis. The primary limitation of the current registration-based approach is that the synthetic image domain (geometrically deformed normal CT) differs from the real sacralization image domain (true bony fusion with altered trabecular and cortical architecture). Unpaired image-to-image translation using a CycleGAN framework [16] offers a principled approach to bridge this domain gap: a generator trained on the 7 real sacralization cases would learn to translate synthetically deformed CT intensities toward the appearance distribution of real sacralization, without requiring paired data. This would substantially improve the realism of synthetic sacralization cases beyond what geometric registration alone can achieve.

TotalSegmentator pseudo-labeling for expanded pelvic supervision. The matched-only synthesis ceiling (23 cases/epoch) is a direct consequence of pelvic ground truth availability. TotalSegmentator [7] segments sacrum and hips using morphological priors rather than vertebra counting and will correctly identify the sacrum and hip structures on all 784 COLONOG cases, including sacralization cases where the superior sacrum incorporates the fused L5. Running TotalSegmentator pseudo-labeling on the full 784-case cohort would provide sacrum and hip supervision for all normal training cases (510 cases instead of 23), expanding the sacralization synthesis pool by $22\times$ and eliminating the matched-only restriction. Pseudo-label quality should be verified against the 33 manually annotated matched cases before adoption.

Castellvi subtype classification. A second-stage model operating on the SpineSurg-CT segmentation output could extract transverse process height and sacral contact distance to classify Castellvi Types I–IV. The spatial segmentation mask produced by SpineSurg-CT provides the anatomical localization necessary for this morphometric analysis.

External validation and prospective study. Validation on independent clinical CT cohorts, including pre-operative spine surgery CTs with surgeon-confirmed vertebral level counts, is the critical next step toward clinical deployment.

5 Conclusion

SpineSurg-CT is the first CT segmentation model with LSTV awareness built into its architecture. It is the first to provide: (1) a voxel-level 3D segmentation mask for the supernumerary L6 vertebra under direct supervised training; (2) calibrated probabilistic LSTV subtype classification (Normal/Lumbarization/Sacralization) from learned morphological features embedded in the segmentation bottleneck; and (3) both outputs simultaneously in a single end-to-end inference pass from raw CT, without upstream vertebra localization or counting steps. Existing tools partition these capabilities without unifying them: TotalSegmentator segments without LSTV awareness; VERIDAH labels without segmenting and requires an upstream segmentation model; Kwak et al. classify without spatial output and require a separate imaging modality. The registration-based sacralization synthesis pipeline — performing ANTs SyN field computation and application entirely in a shared preprocessed coordinate space and restricting synthesis to pelvic-masked reference normals — produces the first reported simultaneous image-and-label sacralization augmentation for CT segmentation training. Together these contributions represent a clinically meaningful advance for preventing wrong-level spine surgery in patients with this anatomically challenging and surgically consequential condition.

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Data and Code Availability

CTSpine1K and CTPelvic1K are publicly available. Model weights, inference code, train/val/test splits, and the CLAIM 2024 checklist will be released at `[GitHubURL---insertbeforesubmission]` under the Apache 2.0 license upon publication.

Figure Captions

Figure 1. Representative SpineSurg-CT segmentation output on a lumbarization test case. Top row: axial slices at three levels showing the predicted 10-class label overlay (L1–blue, L2–cyan, L3–teal, L4–green, L5–lime, **L6–red**, sacrum–orange). Bottom row: corresponding ground truth. The supernumerary L6 vertebra (red) is correctly identified between L5 and the sacrum; TotalSegmentator produces no L6 prediction on this case. **[[Insert figure from inference visualization]]**

Figure 2. Registration-based sacralization synthesis pipeline. Left column: original matched normal spine (CT image and label overlay). Centre column: synthetic sacralization after ANTs SyN field application in preprocessed 256^3 unit space; note the collapsed L5–S1 disc space, enlarged transverse processes, and fused L5+sacrum label (orange). Right column: a real sacralization training case for comparison. Registration and field application are performed entirely in the shared PIR 256^3 1 mm isotropic coordinate space, eliminating the coordinate frame mismatch inherent in native NIfTI-space registration pipelines. **[[Insert from augmentation QC visuals]]**

References

[1] Stephen Albano, Nolan J Brown, Zach Pennington, Andrew Nguyen, Timothy I Hsu, Martin H Pham, and Michael Y Oh. Risks associated with surgical management of lumbosacral transitional vertebrae: Systematic review of surgical considerations and illustrative case. *World Neurosurgery*, 186:e54–e64, 2024.

[2] Michael Karsy and Stephen C Harward. Preventing wrong-level spine surgery. In *Complications in Neurosurgery II*. StatPearls Publishing, 2024.

[3] Tomohiro Ishida, Tetsuhiko Niimura, Tetsuro Sekiya, and Hiroaki Mihara. Lumbosacral transitional vertebrae cause spinal level misconception in surgeries for degenerative lumbar spine disorders. *Asian Spine Journal*, 11(4):518–524, 2017.

[4] Anthony Apazidis, Pedro A Ricart, Christopher M Diefenbach, and Jeffrey M Spivak. The prevalence of transitional vertebrae in the lumbar spine. *Spine Journal*, 11(9):858–862, 2011.

[5] Lorenzo Nardo, Hamza Alizai, Warapat Virayavanich, Fang Liu, Alejandro Hernandez, James A Lynch, Michael C Nevitt, Charles E McCulloch, Nancy E Lane, and Thomas M Link. Lumbosacral transitional vertebrae: association with low back pain. *Radiology*, 265(2):497–503, 2012.

[6] Jaakko Hanhivaara, Juhani H Määttä, Pietari Kinnunen, Jaakko Niinimäki, and Mika T Nevalainen. Castellvi classification of lumbosacral transitional vertebrae: comparison between conventional radiography, ct, and mri. *Acta Radiologica*, 65(1):71–78, 2024.

[7] Jakob Wasserthal, Hanns-Christian Breit, Manfred T Meyer, Maurice Pradella, Daniel Hinck, Alexander W Sauter, Tobias Heye, Daniel T Boll, Joshy Cyriac, Shan Yang, Michael Bach, and Martin Segeroth. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023.

[8] Hendrik Möller, Hanna Schoen, Robert Graf, Matan Atad, Nathan Molinier, Anjany Sekuboyina, Bettina K Budai, Fabian Bamberg, Steffen Ringhof, Christopher Schlett, Tobias Pischon, Thoralf Niendorf, Josua A Decker, Marc-André Weber, Bjoern Menze, Daniel Rueckert, and Jan S Kirschke. VERIDAH: Solving enumeration anomaly aware vertebra labeling across imaging sequences. *arXiv preprint arXiv:2601.14066*, 2026.

[9] David Schinz, Hendrik Möller, Constanze Ramschütz, Paul Platzek, Robert Graf, Anjany Sekuboyina, Malek El Husseini, Matthias Stefan May, Michael Uder, Claus Zimmer, and Jan Stefan Kirschke. A vertebral shape-based classification for labeling the thoracolumbar junction on CT imaging. *European Spine Journal*, 2025.

[10] Jae Hyun Kwak et al. Automated detection of lumbosacral transitional vertebrae on plain lumbar radiographs using a deep learning model. *Journal of Clinical Medicine*, 14(21):7671, 2025.

[11] Yunhe Li, Chunfeng Lian, Fenqiang Zhao, et al. CTSpineSegNet: A 3d spine segmentation network for CT images with cross-scale feature fusion and linear self-attention. *Scientific Reports*, 15(1):1–14, 2025.

[12] Ameya S Tejani, Alexei Klochikhin, Hatem Kalbakdij, et al. Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence*, 6(2):e240300, 2024.

[13] Deng Li, Hans Johnson, et al. CTSpine1K: A large-scale dataset for spinal vertebrae segmentation in computed tomography. *arXiv preprint arXiv:2105.14711*, 2021.

[14] Pengbo Liu, Hu Han, Yuanqi Du, Heqin Zhu, Yinhao Li, Feng Gu, Honghu Xiao, Jun Li, Chunpeng Zhao, Li Xiao, Xinbao Wu, and S Kevin Zhou. CTPelvic1K: A large-scale benchmark for coxal bone segmentation in CT images. *IEEE Transactions on Medical Imaging*, 2021.

[15] Brian B Avants, Nicholas J Tustison, Gang Song, Philip A Cook, Arno Klein, and James C Gee. A reproducible evaluation of ANTs similarity metric performance in brain image registration. *NeuroImage*, 54(3):2033–2044, 2011.

[16] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2223–2232, 2017.